

Isabel Domínguez Jiménez, PhD

Data Scientist • Physicist • Machine Learning Researcher

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Profile

Physicist and data scientist with 12+ years of research experience in high-energy physics at international collaborations (CERN-ALICE, Belle II, NICA). Expert in large-scale statistical analysis, Monte Carlo simulations, and machine learning applied to complex, high-dimensional datasets. Recognized as SNI Level II researcher by CONAHCyT. Complemented scientific background with an industry-focused Data Science Bootcamp (2024). Passionate about transforming raw data into actionable insights and building robust ML pipelines.

Technical Skills

- **Machine Learning & AI:** Scikit-learn, TensorFlow, PyTorch, XGBoost, Graph Neural Networks (GNN), Neural Networks, Logistic Regression, Decision Trees, Random Forests, Monte Carlo methods, Bayesian statistics, optimization algorithms.
- **Data Science:** Pandas, NumPy, SciPy, Matplotlib, Seaborn, Jupyter Notebooks, statistical modeling, data preprocessing, feature engineering, predictive modeling, data visualization (Tableau).
- **Programming:** Python, C++ (ROOT-CERN framework), Fortran, Bash/Shell scripting.
- **Scientific Computing:** ROOT-CERN, GEANT4, PYTHIA, MADGRAPH (Monte Carlo event generators and detector simulation).
- **Data & DevOps Tools:** SQL, Git/GitHub, VSCode, Linux (Ubuntu, CentOS, RedHat), HPC cluster environments, Google Suite, Slack.
- **Languages:** Spanish (native), English (B2).

Key Projects

- **Graph Neural Network for Particle Identification (Belle II experiment)**
Contributed to the development and validation of a GNN-based flavor tagger at the Belle II experiment (KEK, Japan). The model improved CP-violation measurement precision. Published in *Phys. Rev. D* (2024).
- **Neural Network for Beam Parameter Reconstruction (SuperKEKB)**
Co-developed a neural network to reconstruct accelerator beam parameters from beamstrahlung signals, replacing traditional analytic methods. Published in *Nuclear Instruments and Methods A* (2022).
- **Monte Carlo Simulation Pipeline for Heavy-Ion Collisions**
Designed and optimized large-scale C++ simulation pipelines for quark-gluon plasma studies, processing multi-TB datasets from CERN-ALICE and MPD-NICA experiments.
- **Data Science Portfolio (TripleTen Bootcamp, 2024)**
End-to-end projects covering data preprocessing, EDA, predictive modeling, and visualization with Python. Repository: github.com/isadoji/proyectos.
- **Aquaculture Production Optimization (CONAHCyT interdisciplinary project)**
Applied control engineering and optimization algorithms to maximize biomass production in aquatic farms. Awarded **2nd place at ANUIES 2019**.

Professional Experience

Full-Time Professor and Researcher

Autonomous University of Sinaloa (UAS), 2013–2025

Key Achievements:

- Led data analysis and simulation research within ALICE (CERN), Belle II (KEK), and MPD-NICA (JINR) international collaborations, co-authoring 60+ peer-reviewed publications.
- Designed and taught 50+ courses including Computational Physics, Monte Carlo Simulation, Data Science, Optimization, and Scientific Computing at BSc, MSc, and PhD level.
- Supervised 11 theses (BSc, MSc, PhD) on topics including ML for particle identification, Monte Carlo simulation, and data analysis in high-energy experiments.
- Recognized as **SNI Level II** researcher by CONAHCyT (2023–2027), the highest Mexican national research distinction below emeritus.
- Coordinated the interdisciplinary academic group “Decision Support Systems” (Consolidated Level, PRODEP 2022–2027).

Visiting Professor / Sabbatical Researcher

Faculty of Sciences, University of Colima, 2022–2023

Key Achievements:

- Built and benchmarked ML classification models (Logistic Regression, Decision Trees, Neural Networks) in Python for particle physics data analysis.
- Developed data analysis workflows applied to high-energy heavy-ion collision datasets from Belle II and NICA experiments.

Postdoctoral Researcher

Nuclear Sciences Institute, UNAM, 2011–2012

Key Achievements:

- Optimized C++ Monte Carlo simulations for energy-loss modeling in quark-gluon plasma; results published in *Physical Review C*.
- Contributed to CONACyT retention project on quark-gluon plasma energy-loss models (Proj. 128534).

Research Fellow (CERN-ALICE, HELEN Project)

CERN, Geneva, Switzerland, 2006–2009

Key Achievements:

- Analyzed multi-TB experimental datasets from the ALICE detector at the LHC using C++ and ROOT, studying jet quenching and azimuthal correlations in Pb-Pb collisions.
- Developed azimuthal-correlation analysis code later adopted by the ALICE collaboration for publication in high-impact journals.

Education

Bootcamp in Data Science

TripleTen, 2024

Hands-on industry training in data preprocessing, EDA, predictive modeling, SQL, and visualization using Python.

PhD in Physics

National Autonomous University of Mexico (UNAM), 2011

Dissertation: *Jet structure in Pb-Pb collisions at LHC energies*. Advisors: Dr. Guy Paic & Dr. Eleazar Cuautle.

Focus: Monte Carlo simulations, large-scale data analysis, high-energy particle physics.

Master's in Physics

CINVESTAV, Mérida, 2003

Dissertation: *Jet quenching study in heavy-ion collisions*.

Bachelor's in Physics

Autonomous University of the State of Morelos, 2001

Awards & Recognition

National Research System (SNI) – CONAHCyT, Level II

2023–2027

Top-tier recognition for research productivity and impact in Mexico.

Desirable Profile for Full-Time Professors (PRODEP)

2023–2026

Academic Group “Decision Support Systems,” Consolidated Level

2022–2027

2nd Place, ANUIES Innovation Competition

2019

Project: *Optimization of aquaculture farm production via control engineering.*

Honorable Mention, Poster Session, LXI National Physics Congress

2018

Project: *Numerical approximation to Lagrangian mechanics: orbits and scattering.*

Selected Publications

- “A new graph-neural-network flavor tagger for Belle II and measurement of $\sin(2\phi)$.” *Belle II Collaboration. Phys. Rev. D* 110, 012001 (2024).
- “A Neural Network approach to reconstructing SuperKEKB beam parameters from beamstrahlung.” S. Di Carlo, . . . , I. Domínguez, et al. *Nucl. Instr. Methods A* 1042, 167453 (2022).
- “The conceptual design of the miniBeBe detector proposed for NICA-MPD.” *MexNICA Collaboration. JINST* 16, P02002 (2021).
- “Particle physics phenomenology with Python 3.” I. Domínguez Jiménez et al. *Revista Mexicana de Física E* 17(2), 150–155 (2020).
- “Monte Carlo approach for hadron azimuthal correlations in high energy collisions.” A. Ayala, I. Domínguez, et al. *Phys. Rev. C* 86, 034901 (2012).
- Full publication list (60+ papers): [Google Scholar profile](#).

International Collaborations

- **ALICE experiment**, LHC-CERN, Geneva, Switzerland (2006–2014)
- **Belle II experiment**, KEK, Tsukuba, Japan (2013–present)
- **MPD-NICA experiment**, JINR, Dubna, Russia (2016–present)
- **MexNICA Collaboration** – CONACyT (2016–present)